How-To Library

Informatica

Connecting to an HDFS Cluster from Informatica Vibe Data Stream for Machine Data 2.1.0

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Abstract

To write data to a Hadoop Distributed File System (HDFS), you can use the built-in HDFS target service in Informatica Vibe Data Stream for Machine Data. This article provides the steps that you can use to connect to a Hadoop cluster from Vibe Data Stream and write data to the cluster.

Supported Versions

- Vibe Data Stream for Machine Data 2.1.0

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Overview

Use Informatica Vibe Data Stream for Machine Data (VDS) to collect data from different types of sources, such as event logs and real-time logs, and to write this data to the Hadoop Distributed File System (HDFS) on a Hadoop distribution. To write data to a HDFS or Hadoop cluster, use the built-in HDFS target service when you create and deploy a data flow in VDS.

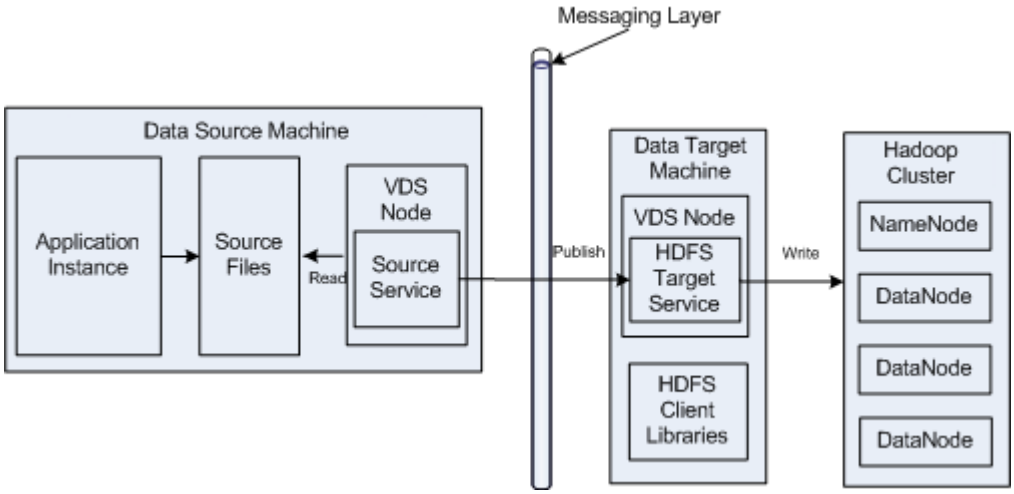
To connect to a Hadoop cluster, you install the client libraries of the HDFS distribution that the Hadoop cluster uses. Install these libraries on the machine on which the VDS Node with the HDFS target service runs. Because VDS does not include HDFS client libraries, you must download and install these libraries.

To connect to a Hadoop cluster and write data, perform the following tasks:

1. Download and install the HDFS client libraries that the Hadoop cluster uses.
2. Verify the connection to the Hadoop cluster from the data target machine.
3. Configure the HADOOPBASEDIR environment variable.
4. Create a data flow with the HDFS target service and deploy the data flow.

Sample Deployment

The following image shows a sample deployment:



In the sample deployment, a source service runs on a VDS Node that runs on the data source machine. The HDFS target service runs on a VDS Node that runs on a data target machine. The source service reads data from the source and publishes the data. The HDFS target service reads the data that the source service publishes and writes the data to the Hadoop cluster. The data target machine is a client machine that is outside the Hadoop cluster. The Hadoop cluster includes multiple machines that comprise the distributed file system. The Hadoop cluster includes the primary NameNode and the DataNodes. The NameNode manages the file system metadata and DataNodes store the data in an HDFS cluster.

Step 1. Install HDFS Client Libraries for the Hadoop Cluster

Download and install the HDFS client libraries for the Hadoop cluster on the machine on which the VDS Node with the HDFS target service runs.

You can use the following Hadoop distributions:

- Cloudera Distribution. See the Cloudera downloads page at the following location: http://www.cloudera.com/content/support/en/downloads.html?productID=Mg0r6damX1
- Apache Hadoop. See the Apache downloads page at the following location: http://archive.apache.org/dist/hadoop/core/
- Hortonworks Data Platform. See the Hortonworks Data Platform page at the following location: http://hortonworks.com/hdp/

Note: Before you download and install the HDFS client libraries for a particular HDFS cluster, verify the version of the Hadoop distribution. The installation steps differ based on the distribution that you use. Follow the installation steps provided in the downloads location.

Step 2. Verify Connection to the Hadoop Cluster

After you install the HDFS client libraries, verify the connection to the Hadoop cluster. Verify the NameNode and port on which the NameNode listens for connections.

To verify the HDFS connection, run the following command from the command prompt on the data target machine:

```
$HADOOP_HOME/bin/hadoop fs -ls hdfs://<namenode-name>[:<port>]/
```

Where

- `HADOOP_HOME` is the Hadoop installation directory.
- `namenode-name` is the host name or IP address of the NameNode.
- `port` is the port number on which the NameNode listens for connections.

Step 3. Configure the HADOOPBASEDIR Environment Variable

VDS uses the HADOOPBASEDIR environment variable to find the HDFS client libraries for the Hadoop cluster. Set the HADOOPBASEDIR environment variable on each VDS Node machine on which you want to run the HDFS target service.

The environment variable must point to the Hadoop base directory. The environment variable is in the following format:

```
HADOOPBASEDIR=<Hadoop_Home_Directory>
```

For example: `HADOOPBASEDIR=/usr/hadoop-2.0.2-alpha`

Step 4. Create and Deploy a Data Flow

To connect to the Hadoop cluster, use the Informatica Administrator for VDS (the Administrator tool) to create and deploy a data flow with a **File** source service and an **HDFS** target service.

To connect to the Hadoop cluster, perform the following tasks:

1. Create the data flow.
2. Add the source service and target service to the data flow.
3. Map the source service and target service to VDS Nodes.
4. Deploy the data flow.

Creating the Data Flow

Use the Administrator tool to create the data flow.

1. Log in to the Administrator tool.
2. On the **Domain** tab, click **Services and Nodes**.
3. In the **Domain Navigator** pane, select the **Vibe Data Stream** Service.
4. In the **Data Flows** pane, right-click **All Data Flows**.
5. Specify the properties of the data flow and click **Save**.

Adding the Source Service and Target Service to the Data Flow

After you create the data flow, add the source service and target service to the data flow.

1. In the **Data Flows** pane, click the data flow to which you want to add a source.
2. From the **Entity Types** pane, drag the **File** source service to the **Data Flow Designer** pane.

   The **New Source** dialog box appears.
3. Specify the properties of the source service and click **OK**.
4. From the **Entity Types** pane, drag the **HDFS** target service to the **Data Flow Designer** pane.

   The **New Target** dialog box appears.

5. Configure the following properties for the HDFS target service type:

Property	Description
Entity Name	Name of the HDFS target service. Maximum length is 32 characters.
Destination	URI of the target file to which to write data. The HDFS target service type supports the following URI formats: **HDFS URI format** `hdfs://<namenode-name>[:<port>]/<path>/<file-name>` Where - `namenode-name` is the host name or IP address of the HDFS NameNode. - `port` is the port number on which the HDFS NameNode listens for connections. You can omit the port number if you have configured HDFS to listen for connections on the default port, 8020. - `path` and `file-name` represent the location of the target file in the target file system. The URI format is suitable for a standalone HDFS target service. The URI is also suitable for an HDFS target service that runs on a node that is not part of a high-availability setup. To use multiple target service instances for load balancing or high availability, use variables in the URI. **Note:** The destination URI must be the same URI that you used to verify the connection to the HDFS cluster.
Rollover Count	Limit for the number of files that can exist on the target at any particular time. Default is 1024.
Rollover Size	Target file size, in gigabytes (GB), at which to trigger rollover. Default is 1. A value of zero (0) means that the HDFS target service does not perform rollover based on size.
Rollover Time	Length of time, in hours, to keep a target file active. After the time period has elapsed, the target service rolls the file over. Default is 0. A value of zero (0) means that the HDFS target service does not perform rollover based on time.
Force Synchronization	Flush the client's buffer to the disk device every 1 second. If you enable forceful synchronization, data written by the target service is visible to other readers immediately. Forceful synchronization degrades the performance of VDS. For more information about forceful synchronization, see the Hadoop documentation. Default behavior is to not synchronize forcefully.
UM XML Configuration	Specify the UM configurations that the target service uses. Maximum length is 1000 characters.

Mapping the Source Service and Target Service to Vibe Data Stream Nodes

To deploy a source service or target service on a particular VDS Node, map the service to that node. You can map a service to multiple VDS Nodes. You can map services to newly started VDS Nodes and to VDS Nodes that are running services. You can also map a service to a VDS Node that you have not yet started. When the VDS Node starts, it downloads its configuration from Apache ZooKeeper and starts the service.

1. In the Administrator tool, on the **Domain** tab, click **Services and Nodes**.
2. In the **Domain Navigator** pane, select **Vibe Data Stream Service**.

3. In the upper-right corner of the **Vibe Data Stream Service** pane, click **Vibe Data Stream**.
4. In the **Data Flow Designer** pane, select the source service or target service to which you want to map nodes.
5. In the **Entity Details** pane that appears, click the **Nodes** tab.
6. Click **Add new record**.
7. In the **Node Name** column, in the box that appears, enter the name of the node on which you want to run the service.
8. Press **Enter**.
9. If you want to deploy the service on multiple nodes, repeat steps 6 through 8 until you have added all the nodes you want.
10. Click **Save**.

Deploying a Data Flow

Deploy a data flow to start the source services and target services in the data flow.

1. In the Administrator tool, on the **Domain** tab, click **Services and Nodes**.
2. In the **Domain Navigator** pane, select **Vibe Data Stream Service**.
3. In the upper-right corner of the Vibe Data Stream Service pane, click **Vibe Data Stream**.
4. In the **Data Flows** pane, select the data flow that you want to deploy.
5. In the **Data Flow Designer** pane, click **Deploy**.

   VDS deploys the data flow, and a check mark appears beside the name of the data flow in the **Data Flows** pane. The check mark indicates that you have deployed the data flow.

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